FifthWave Metals

PREDICT. OPTIMIZE. DISCOVER.

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PRIVATE AND CONFIDENTIAL

COPPER SUPPLY IS BREAKING.

~20 Mt

Annual new copper supply needed by 2035

Demand is rising. Production is not.

0.6%

Average Cu ore grade today (down from 1.5% in the 1980s)

33% more throughput for the same metal.

~28 yrs

Projected discovery-to-production timeline (2025 onward)

Up from 11 years in the 1990s.

The industry is making bigger bets on worse ore with less time to get it right.

And the current toolkit was built for a world of higher grades and shorter timelines.

More capital at risk

~$300B+ required to close the projected 2035 supply gap

Less room for error

Mistakes compound over decade-long build cycles

Novel chemistry required

Declining grades force a shift to leaching routes

Sources: IEA Global Critical Minerals Outlook 2025, S&P Global ore grade data, BHP, Wood Mackenzie.

BROKEN FLOWSHEETS ARE THE PROBLEM. AND CURRENT POINT SOLUTIONS ARE NOT THE ANSWER.

Everyone is Throwing Darts

The market is flooded with point solutions - individual chemistries, sensors, and heating technologies. None solve at the system level.

- New sensors (muons, ERT, DTS) ✕
- Novel leach chemistries ✕
- Heating technologies ✕
- Oxidant variants ✕

Because these technologies are applied in isolation.

The Nickel Laterite Warning

For 50 years, nickel laterites tried one flowsheet after another - HPAL, electro-RKFs, heap leaching, atmospheric leaching. Each generation reinvented the process. Each one broke.

$35B+[1]

- Twelve deposits.
- Eight different flowsheets.
- All shut down or failed economically for first owners.
- The flowsheets failed, not the geology.

Copper is now on the same path.

Declining ore grades force a shift to leaching. Without a solution we will repeat nickel's 50-year cycle of broken flowsheets.

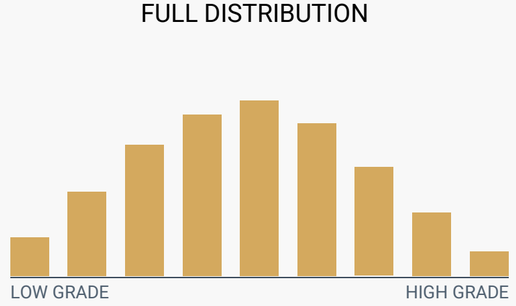
Source: IEA GCMO 2025.
(1) Internal Analysis of nickel laterite projects – Capital Escalated to 2026 dollars, Bulong, Cawse, Murrin Murrin, Ravensthorpe, Barro Alto, Ambatovy, Goro, Koniambo, CMSA, Ramu, Yabulu, Nonoc.

MINING IS FRAGMENTED. BILLIONS ARE LOST.

Four teams. Four optimisations. Zero system view.

Resource Geologist

"What is the size and grade of this resource?"



IGNORES

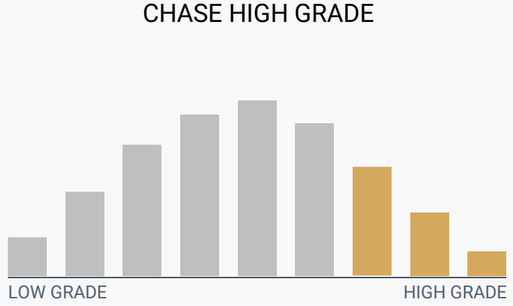
- Ore hardness variability
- Geometallurgical response
- Mill feed implications

FOCUSES ON

- Grades & tonnes
- Resource confidence
- JORC classification

Mining Engineer

"How do I always mine the highest grade first?"



IGNORES

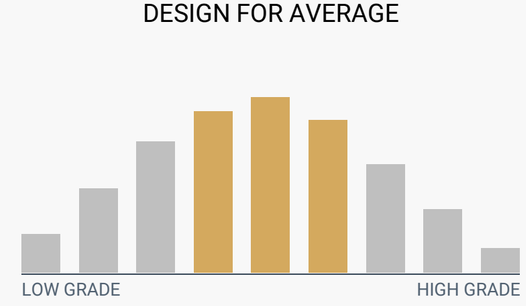
- What the plant receives
- Feed hardness & mineralogy
- Mine–plant interaction

FOCUSES ON

- High-grade zones
- Extraction sequence & NPV
- Equipment utilisation

Process Engineer / EPCM

"What is the average ore the plant will see?"



IGNORES

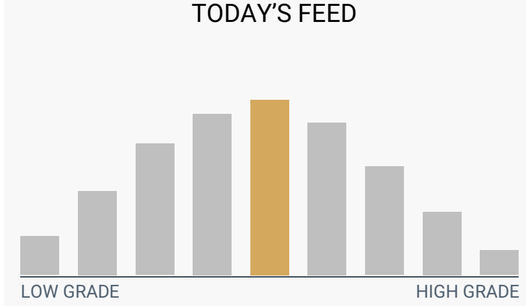
- Real ore variability
- Off-design feed performance
- Operational flexibility needed

FOCUSES ON

- Average (P50) feed grade
- Test work composite
- Equipment sizing & capex

Plant Metallurgist

"How do I process the ore I am given?"



IGNORES

- Why feed is what it is
- Upstream mine decisions
- Proactive optimisation

FOCUSES ON

- Today's feed conditions
- Reagent regime & recovery
- Throughput & uptime KPIs

Source: Fifthwave Metals analysis.

WE ARE SOLVING THE MISSING MIDDLE.

No incumbent owns prediction between geology and engineering.

Upstream ↑

Mine Planning / Geological

DATAMINE · Deswik · GEOVIA · SEEQUENT

Model geometry, not chemistry. Grade ≠ Recovery.

Deterministic Process Simulators

aspentech · CASPEO Beyond process simulation · METSIM INTERNATIONAL

Assume known flowsheet. No uncertainty quantification.

Gap

THE SOLUTION

A PROBABILISTIC, AI-PHYSICS-CONSTRAINED, CONTINUOUSLY LEARNING INFRASTRUCTURE

Downstream ↓

Automation & Consultants

FLUOR · HATCH · Honeywell

Project-specific, non-compounding.

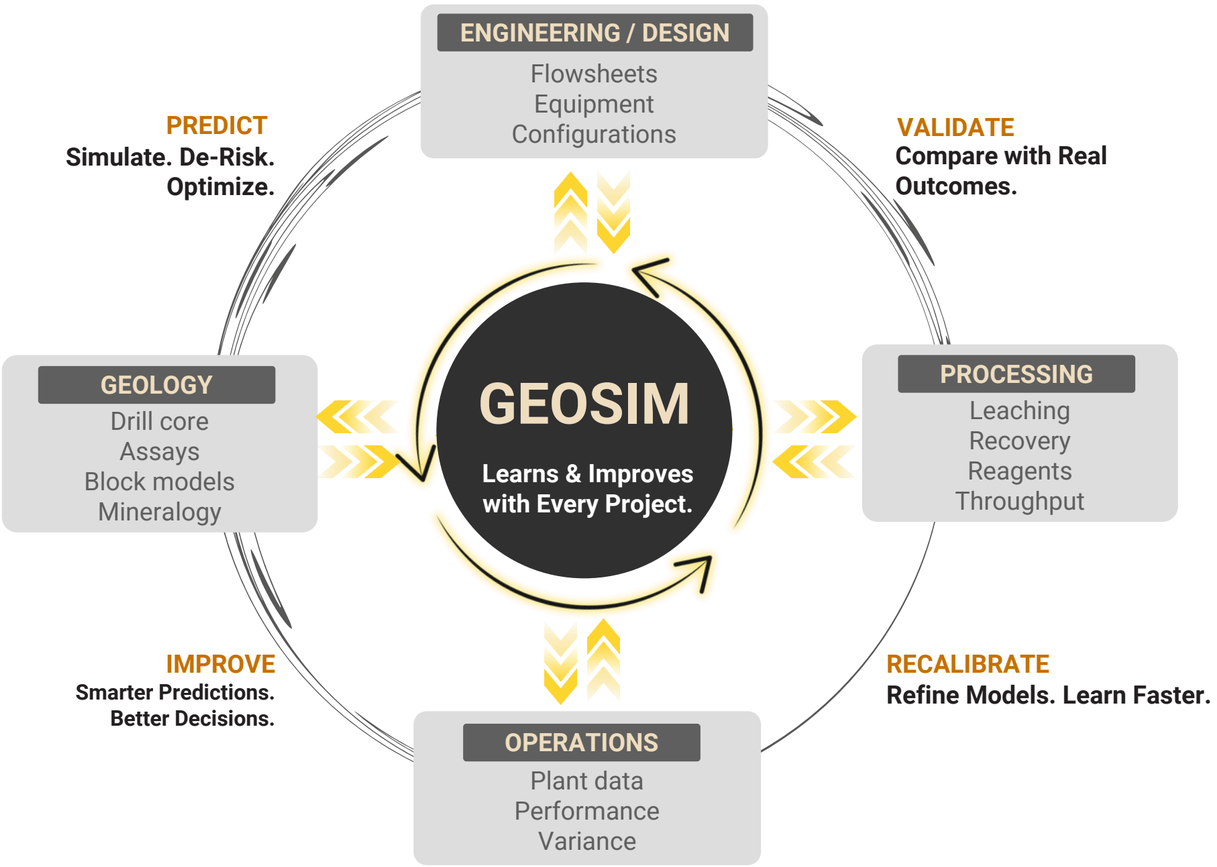
Digital Twins & Advanced Process Controls

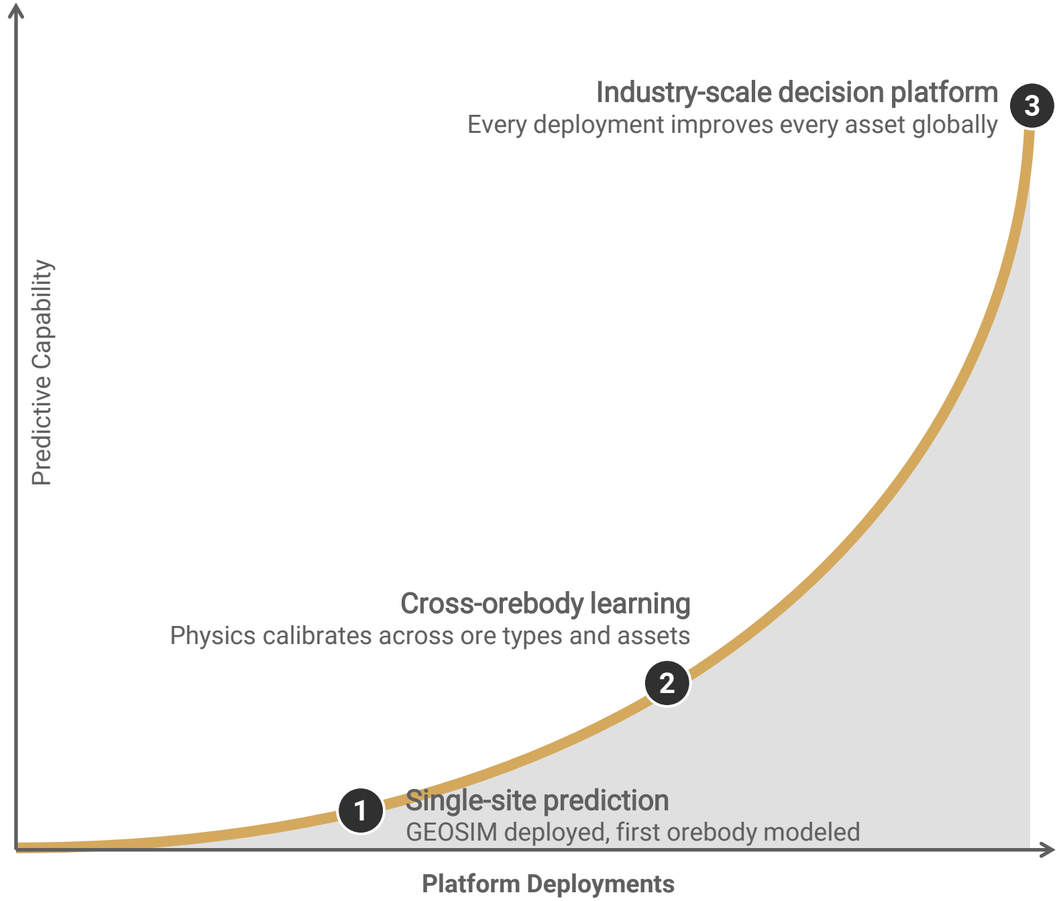
ANDRITZ · FLSMIDTH · Metso

Require live plant data. Can't model what doesn't exist.

Source: Fifthwave Metals analysis.

A PLATFORM BUILT ON PHYSICS NOT FRAGMENTED DATA.





BREAKS DOWN SILOS	REDUCES RISK	COMPOUNDS INTELLIGENCE	BUILDS A BETTER FUTURE
Connects what's been disconnected.	Test more. Fail less. Deliver with confidence.	Every project makes the next one smarter.	More recovery. Less waste. Stronger outcomes.

Every deployment improves every future decision, across all assets.

Source: Fifthwave Metals analysis.

THE SOLUTION: GEOSIM

Drill core assays

- *Grade*
- *Multi-element geochemistry*

Resource Models & Geological block model

- *Domain boundaries*
- *Ore variability*

Mineralogy (QEMSCAN/MLA)

- *Mineral associations*
- *Liberation*

Testwork results

Column tests / Bottle rolls / Pilot

INPUT

AI-PHYSICS ENGINE

Thermodynamics, kinetics

CFD (fluid flow), DEM (particle mechanics), FEM (stress and heat transfer)

Flowsheet simulation

Ensemble Kalman state estimation

Value of Information / Bayesian Uncertainty Quantification

Optimization and Scheduling

INTEGRATED DATA LAYER

- Geological data
- Mineralogy
- Lab test work
- Process data
- Sensor network
- Outcome actuals

FIFTHWAVE METALS' CORE PROPRIETARY MODELS

OUTPUT

Resource Knowledge

Live resource model inc

Mine Plan and Schedule

Probabilistic LOM, 5-year plan, monthly schedule, daily execution

Engineering Construction and Capital Allocation

Pre-construction

Processing Plant

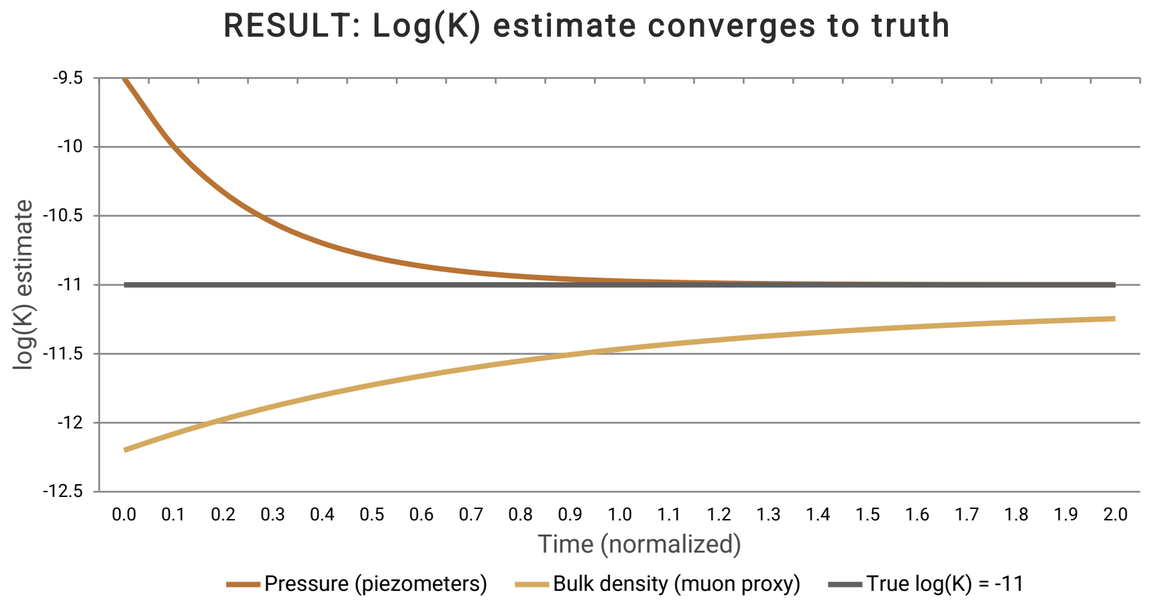
- *Recovery*
- *Throughput*
- *Costs*

WE HAVE DEMONSTRATED THAT OUR APPROACH LEARNS THE SYSTEM.

GEOSIM combines physical models with real observations to reconstruct system behavior and predict outcomes under uncertainty. This approach can be used across the mining value chain.

VALIDATION SETUP

1. We have already demonstrated this capability
2. Started from incorrect assumptions
3. Using only noisy observations
4. Converged to the true system performance
5. Which allows new and better decisions to be made in real time



Example to demonstrate approach – Predicting Heap Permeability from noisy sensors (synthetic case)

Without GEOSIM, decisions based on incorrect assumptions.
Hidden bias persists

With GEOSIM, system converges to true behavior. Decisions update as reality is learned

WHAT THIS PROVES: This demonstrates a general capability to learn true system behaviour from noisy operational data, enabling continuous improvement across the mining value chain. **The next step is validation against live operational data before scaling.**

Source: Persephone Resources / FifthWave Metals internal validation, 2026.

WHERE WE ARE TODAY. TRACTION ACROSS THE FUNNEL.

Technical validation complete. Commercial pipeline building across all customer tiers.

COMPLETE

TECHNICAL VALIDATION

PHASE 1

complete

Engine converges in controlled tests. Model recovers unknown heap permeability from synthetic sensor data.

ENGAGED

ACTIVE TECHNICAL ENGAGEMENT

6

operators in technical discussions

1 Tier-1 operator
3 mid-tier operators
2 emerging producers

PROGRESSING TO PILOT

COMMERCIAL CONVERSATIONS

1

mid-tier operator in active pilot scoping

First paid pilot negotiation underway.

FUNDED BY SEED

PAID PILOTS

FIRST PILOTS YEAR 1

Seed funds 1-2 paid pilots. Target: validate predictions vs. column test outcomes on real ore bodies by month 18.

NEXT 18 MONTHS

- 2 PAID PILOTS SIGNED
- 1 PILOT DELIVERED
- VALIDATION VS. REAL OUTCOMES
- PROVISIONAL PATENTS FILED
- SERIES A POSITIONING

EVERY PLAYER IN MINING NEEDS THIS.

Four Buyer Types. One Universal Problem.

MINING MAJORS

BHP · Newmont · RioTinto · Teck

Portfolio-level capital allocation across 10-50+ sites. Evaluating new flowsheets for declining-grade assets.

Enterprise Contracts

MID-TIER & JUNIOR MINERS

CAPSTONE COPPER · HUD · IVANHOE MINES · SOLARIS RESOURCES

De-risk single flagship projects before committing $500M+ capex. Optimize ramp-up and commissioning. $300-500K/site/yr.

Site Contracts

EXTRACTION INNOVATORS

Ceibo · endolith · JETTI · STILLBRIGHT

Every novel leach chemistry needs GEOSIM's physics layer to predict how it behaves across ore variability before scaling from lab to heap.

Seat + Simulation Credit Pricing

SOFTWARE-FIRST OPERATORS

KoBold Metals · MARIANA MINERALS · nuton (A Rio Tinto venture)

These companies deploy capital across multiple sites. They need a pre-operational prediction layer that travels with them, not consultants re-scoped for each project.

Portfolio Licensing

CUSTOMER IP PROTECTION:
Customer flowsheets, orebody data, and outcomes stay within their own tenant. FifthWave does not use customer IP to build competing chemistry.

Source: Fifthwave Metals analysis.

A SOFTWARE BUSINESS TODAY. A PLATFORM BUSINESS TOMORROW.

SOFTWARE

TODAY

Physics-based decision layer

$150M

Bottom-up SAM, copper hydromet

- ~375 addressable hydromet sites globally
- $300-500K ACV per site, $2M per enterprise
- Comp: Seequent acquired for $1.05B (2021)

Year 5 ARR target: $50-75M

PLATFORM + CROSS-SITE IP

5-10 YRS

Software + proprietary physics models

$1B+

Category-leader potential

- Cross-site learnings generate proprietary physics models
- Those models unlock novel flowsheets no operator could derive
- We license our IP; customer flowsheets remain private

SOFTWARE DEPLOYMENTS → CROSS-SITE LEARNINGS COMPOUND → FIFTHWAVE-OWNED IP

HOW FIFTHWAVE METALS WINS.

Four moats. Each compounds with every deployment.

01

PHYSICS

**AI is a commodity.
Physics is not.**

Why incumbents can't copy:

- Aspentech runs deterministic simulations on operating plants.
- GeoSim is probabilistic, built for pre-construction ore bodies.
- Different math. Different data. Different use case.

25+ years

of thermodynamic and kinetic modeling in the core engine

02

POSITION

**We occupy a window
no incumbent owns.**

Why the gap is structural:

- Upstream tools (Seequent, Datamine) stop at geometry.
- Downstream tools (Aspentech, Metso) need live plant data.
- Neither has the stack or the incentive to fill the middle.

0 incumbents

in the pre-construction hydromet decision window today

03

DATA

**Every deployment widens
the gap.**

Why a fast follower fails:

- Each pilot generates paired orebody + outcome data we own.
- A new entrant starts at zero sites.
- By our 10th deployment, catching up takes a decade.

Asymmetric

outcome dataset compounds with every deployment

04

TRUST

Built by miners, for miners.

Why this matters:

- Operator trust and mining experience valued.
- Grant: 25+ years at BHP. Atusa: Shell + operator experience.
- This is what unlocks procurement at Tier-1 majors.

$15B+

in capital decisions led by the founding team

A moat that grows with every deployment.

38+ YEARS OF HYDROMETALLURGICAL EXPERTISE.

Atusa Sadeghi

Co-founder & CEO

Mining Engineer, Venture Capitalist

- **13 years** across operations and mining-tech investments
- **VP of Investments at Resource Capital Fund**, investing across METS, critical minerals, and industrial AI
- **Director of Mining Technology at Shell**, screening 1,500+ technologies and executing 30+ strategic partnerships
- **Site Engineer** at Guyana Goldfields (Amazon rainforest), First Bauxite (Guayana), and Codelco (Chuquicamata)

Grant Caffery, PhD

Co-founder & CSO

Hydrometallurgist, leader in mineral extraction innovation

- **25+ years of executive leadership** across Tier-1 global mining organizations
- **VP Head of Innovation at BHP**; directed ~$100M annual capital across 100+ performance initiatives
- **Led $1B heap leach feasibility study** and ~$15B in M&A diligence
- **Board experience** across technology investment, capital allocation, and asset performance governance

$3M SEED RAISE. 18-MONTH RUNWAY TO SERIES A.

Incubated by RACAPITAL PLANETARY HEALTH

Paid pilots prove the physics translates from synthetic to operational.

USE OF FUNDS

40% **Platform R&D & simulation engine**
Technical team + cloud compute

25% **Paid pilot deployments**
Field trials with 1-2 customers

15% **Business development & GTM**
Fractional BD + travel

10% **IP protection & patent filings**
Provisional + full filings

10% **G&A, legal, working capital**
Corporate infrastructure

MILESTONES

M3-6
1-2 paid pilots signed with named customers

M6-9
End-to-end platform demo on real ore body data

M9-15
Pilot validation vs. column test outcomes

M12
Provisional patents

M15-18
Series A raise initiated with validation in hand

TEAM PLAN

CURRENT (2)

Atusa Sadeghi - CEO

Grant Caffery, PhD - CSO

SEED HIRES (4)

Senior Software Engineer (full-time)

Computational physicist (full-time)

Field Engineer (phased - pilot deployments)

Fractional BD (phased - mining industry network)

THANK YOU

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